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7 May 2026

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7.7 Pre-Quiz: Logarithms (no calculator)

1. Write each of the following expressions in the form $\ln k$, where $k \in \mathbb{Z}^+$.

   (a) $\ln 3 + \ln 4$

   (b) $3 \ln 2$

   (c) $-\ln \frac{1}{2}$

2. Let $\log_{10} 2 = p$ and $\log_{10} 3 = q$.

   (a) Find an expression for $\log_{10} 24$ in terms of p and q.

   (b) Find an expression for $\log_3 8$ in terms of p and q.

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3. It is given that $\log_{10} a = \dfrac{1}{3}$, where $a > 0$.

   Find the value of

   (a) $\log_{10}\left(\dfrac{1}{a}\right)$;

   (b) $\log_{1000} a$.

4. Solve the equation $2\ln x = \ln 9 + 4$. Give your answer in the form $x = p\,e^{q}$ where $p, q \in \mathbb{Z}^{+}$.

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5. Solve the equation $3\log_8(10x) - \log_4 x = 1$, for $x > 0$.

6. The function f is defined as $f(x) = \log_2(8x)$, where $x > 0$.

   (a) Find the value of

      (i) $f(2)$;

      (ii) $f\left(\dfrac{1}{8}\right)$.

   (b) Find an expression for $f^{-1}(x)$.

   (c) Hence, or otherwise, find $f^{-1}(0)$.

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The function f is defined as $f(x) = \log_2(8x)$, where $x > 0$.

The graph of $y = f(4x^2)$ can be obtained by translating and stretching the graph of $y = \log_2 x$.

(d) Describe these two transformations specifying the order in which they are to be applied.